Sujet Projet Neural Network and Deep Learning :

Dear all,

It is now time to put into practice the concepts covered during the course. You will carry out a complete Deep Learning project, from raw data to model development and interpretation of the results.

📌 Project Steps

1. Choice of domain
Identify a topic that interests you, such as health, finance, environment, sport, art, or a topic related to your internship or apprenticeship.

2. Data collection
Find a relevant dataset. You should prioritize open data sources.

Recommended sources include: **INSEE**, **Kaggle**, **UCI Machine Learning Repository**, **Data.gouv.fr** **ou** **OpenDataSoft**.

3. Exploratory Data Analysis (EDA)
Analyze your data in order to understand its structure, identify correlations, detect possible biases, and highlight relevant patterns.

4. Baseline model
Start by implementing a simple baseline model in order to establish a reference performance.

5. Deep Learning architecture
Select and implement one or several Deep Learning architectures adapted to your problem.

The architecture must be relevant to the type of data and the objective of the project.

6. Optimization and improvement
Apply the techniques covered in class to improve your results, such as regularization, data augmentation, transfer learning, hyperparameter optimization, advanced architectures, or better training strategies.

7. Results interpretation
Analyze your results critically. Do not simply report a performance score. Explain what the score means, discuss the strengths and limitations of your model, and draw concrete conclusions

📝 Expected Deliverables

Your work must be submitted no later than **one week after the last session**. The submission must include:

1. Notebook
A clean, well-commented, and executable notebook.

2. Report Word
A synthesis document of **maximum 10 pages**.
Important: the report must not contain code. Voir juste en dessous pour les consignes du rapport

⚖️ Evaluation Criteria — Total: 20 points

The evaluation will focus on scientific rigor and clarity of communication.

Code quality — 5 points
Readability, structure, and comments.

Report quality and analysis — 15 points
This includes:

- A clear definition of the problem.
- The quality of the exploratory analysis and visualizations.
- The relevance of your methodology: Why this model? Why these choices?
- The interpretation of the results: Do not simply report a score. Explain what it means and draw concrete conclusions from it.

Important note:
An excellent score with poor analysis will always be worth less than an average score accompanied by a strong, critical, and well-argued analysis.

Good luck to everyone. Be creative, rigorous, and analytical!